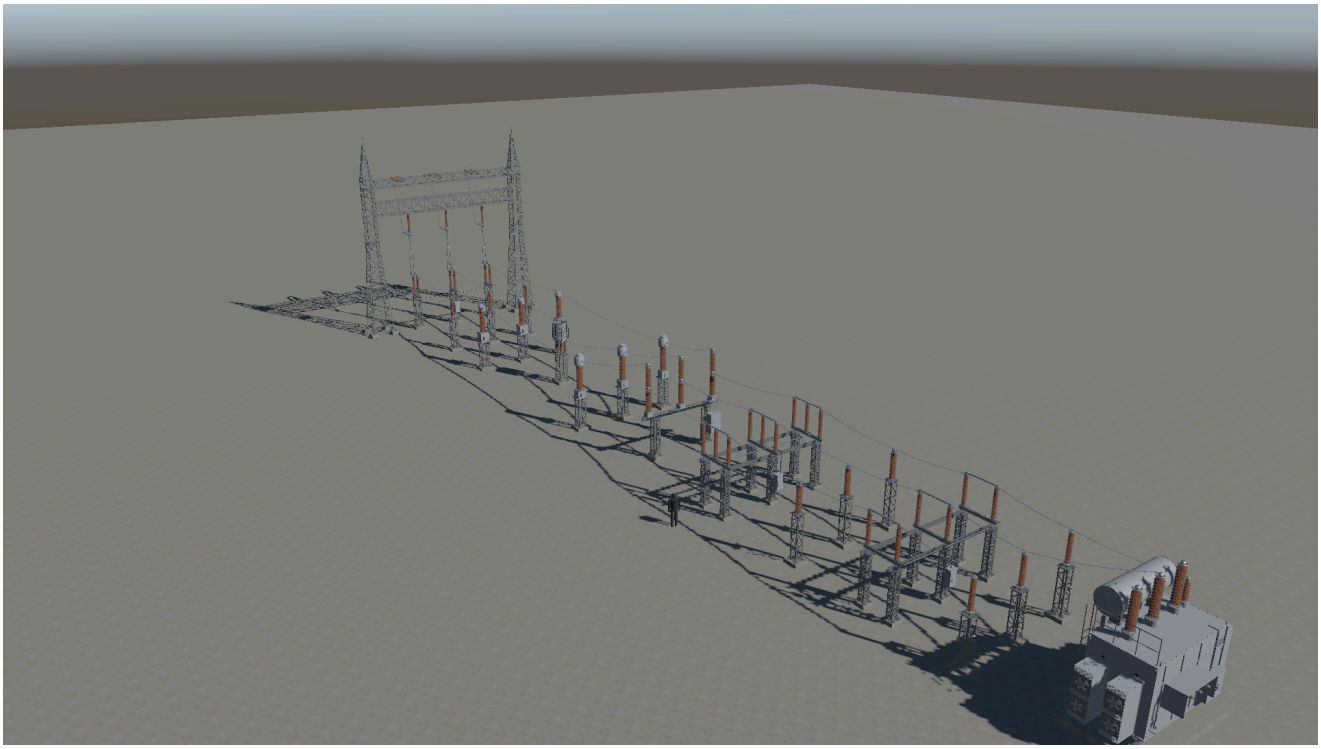
Substation 138 kV Kit — Documentation

Version 1.0.0 · Publisher: Arkeh_3D · Unity 6000.0 LTS or newer · URP and Built-in Render Pipeline



What is in the package

14 prefabs of generic high-voltage substation equipment (138 kV class), modelled at real scale with typical dimensions, on lattice steel supports. They are generic designs, not copies of any manufacturer's product. Every prefab has three LOD levels, primitive colliders, moving parts where the real device has them and connection points on the line terminals.

Prefab	Description	Height (m)	Triangles LOD0 / LOD1 / LOD2	Moving parts	Connection points
Breaker_138kV	Live-tank circuit breaker, 3 poles, on two lattice columns; control cabinet with an opening door.	5.70	59,108 / 29,744 / 12,392	ControlCabinet_Door	6
CurrentTransformer_138kV	Current transformer (single phase, head type) on a lattice pedestal.	4.77	17,916 / 10,380 / 3,932	—	2
Disconnector_CenterBreak_138kV	Center-break disconnector, 3 phases; each phase has two rotating half blades.	4.56	89,968 / 51,684 / 22,340	ControlCabinet_Door, Blade_1A, Blade_1B, Blade_2A, Blade_2B, Blade_3A, Blade_3B	6
Disconnector_DoubleBreak_138kV	Double-break disconnector, 3 phases; a full blade on the rotating center insulator.	4.56	110,532 / 51,648 / 26,472	ControlCabinet_Door, Blade_1, Blade_2, Blade_3	6
Disconnector_SemiPantograph_138kV	Semi-pantograph (vertical-reach) disconnector, 3 phases: arms straight up when closed, folded along the bay when open; suspended fixed contact on a hanging insulator, made to bolt under the Gantry_138kV_SwitchBeam lower beam.	8.98	88,796 / 43,696 / 18,288	LowerArm_1, LowerArm_2, LowerArm_3, ControlCabinet_Door, UpperArm_1, UpperArm_2, UpperArm_3	6
Gantry_138kV	Line entry gantry: two tapered lattice towers with peaks and a lattice beam.	14.88	49,056 / 15,840 / 7,888	—	5

Gantry_138kV_SwitchBeam	Line entry gantry with a second, lower lattice beam that carries the suspended fixed contacts of the semi-pantograph disconnectors.	14.88	60,416 / 20,672 / 10,656	—	5
InsulatorString_138kV_Double	Double suspension insulator string with yoke plates.	2.04	12,828 / 5,460 / 2,900	—	2
InsulatorString_138kV_Single	Suspension insulator string, 10 discs, with shackle, clamp and conductor stub.	1.77	6,612 / 2,788 / 1,508	—	2
LineTrap_138kV	Line trap (air-core coil) on three support insulators and a lattice pedestal.	4.34	29,456 / 16,152 / 5,712	—	2
PostInsulator_138kV	Post insulator with busbar clamp on a lattice pedestal.	4.13	14,460 / 7,756 / 2,764	—	1
PowerTransformer_138kV	Three-phase power transformer 138/13.8 kV: tank, HV and LV bushings, conservator, radiators, control cabinet with door.	5.71	53,844 / 31,912 / 12,968	ControlCabinet_DoorRight, ControlCabinet_DoorLeft	9
SurgeArrester_138kV	Surge arrester (single phase) with insulating base and discharge counter.	4.07	16,900 / 8,244 / 4,148	—	1
VoltageTransformer_138kV	Inductive voltage transformer (single phase) on a lattice pedestal.	4.69	19,300 / 9,348 / 4,228	—	1

Total: 629,192 / 305,324 / 136,196 triangles (LOD0 / LOD1 / LOD2). A 1.75 m scale figure prefab is included for the demo scenes.

Technical details

- **Scale and axes:** 1 unit = 1 m. Y is up and the front of every device faces +Z. The prefab pivot is on the ground at the centre of the base (suspended insulator strings: at the attachment point, hanging towards −Y).
- **LOD:** three levels per prefab, in an LOD Group (screen heights 30 %, 10 % and 1.5 %). LOD1 drops the small bolts and reduces the round segments; LOD2 drops all fasteners and simplifies insulator sheds and lattice bracing.
- **Meshes:** one mesh per material per LOD; moving parts are separate meshes with the pivot on their hinge or rotation axis.
- **Colliders:** primitive only — a box or a vertical capsule per sub-assembly, plus a box on every moving part. You can walk under the gantry beam and between the columns.
- **Materials:** 8 shared materials for the whole kit (galvanized steel, coated steel, aluminium, brown porcelain, grey insulator, black rubber, concrete, labels atlas). URP Lit by default; SRP Batcher and GPU Instancing compatible.
- **Textures:** 24 tileable PNG files (3 per material): albedo, normal map (imported as Normal Map) and metallic map (R = metallic, A = smoothness), 1024 × 1024 per metre (UV in metres); the labels atlas is 256 × 256 (albedo, normal, metallic). All textures are procedural and were made for this kit.

- **Lightmaps:** the models have a second UV set generated at import (Generate Lightmap UVs), so baked lighting works; the first UV set is in metres and overlaps by design for tiling.

Render pipelines

- **URP:** works out of the box (the materials use Universal Render Pipeline/Lit).
- **Built-in Render Pipeline:** after importing the package, open Extras/ and import *BuiltInRP_Materials.unitypackage*. It replaces the 8 kit materials with the same materials on the Standard shader (same names and GUIDs, so every prefab keeps working).

Moving parts and connection points

- **OperablePart** (on the pivot of every moving part): call `Open()`, `Close()`, `Toggle()` or `SetOpenAmount(0..1)`. Fields: rotation axis, open angle in degrees, speed in degrees per second, start open. The semi-pantograph disconnector has two parts per phase: UpperArm_n is a child of LowerArm_n, so open both (the lower arm tilts 75° to 15° above horizontal and the upper arm folds 165°

back over it, with the ratio that lowers the head straight off the fixed contact); the crank, link rod and gearbox are static — the drive linkage is not animated.
- **ConnectionPoint** (on every line terminal, named Anchor_*): position plus the outward direction (`localDirection`), to attach your own conductors or busbars.
- **ConductorSpan** (conductor tool): put it on an empty GameObject, assign two ends (ConnectionPoints or any Transforms), the sag and the radius, and it builds the conductor mesh — in the editor, live while you move the ends, and at runtime (`ConductorSpan.Create(a, b, material, sag, radius)`). UVs in metres, so the kit materials tile on it. The curve is the parabolic approximation of the catenary (accurate for sags up to about 10 % of the span).
- **KitDemoController** (demo camera): right mouse to orbit, wheel to zoom, left click on a moving part to open or close it, O opens all, C closes all. It works with the Input System package and with the legacy Input Manager.

Scenes

- **Demo_Bay138kV:** a complete 138 kV line bay in the real order — line entry gantry with the switch beam and double dead-end strings, semi-pantograph line disconnector hanging its fixed contacts from the switch beam, surge arresters, voltage transformers, line trap, current transformers, circuit breaker, double-break bus disconnector and the busbar on post insulators; on the transformer side, a center-break disconnector, surge arresters next to the transformer and the power transformer, with ConductorSpan conductors between the terminals of each phase.
- **Showcase:** every prefab side by side with the scale figure.

Performance (Demo_Bay138kV, initial camera)

552,966 triangles in view, 184 renderers, 7 materials; 1.67 ms per frame (CPU + GPU, Camera.Render in the Unity editor) at 1920 × 1080 (URP, MSAA 4×, soft shadows) on NVIDIA GeForce RTX 4050 Laptop GPU.

Windows player build, same scene and view, 60.0 s: average 1306.55 fps (frame time 0.77 ms, 1 % low 580.24 fps), 1920x1080, vsync off, NVIDIA GeForce RTX 4050 Laptop GPU (measured 2026-10-18).

Content origin

All 3D models are procedural: every mesh is generated by the author's own Python toolchain for Blender, from typical 138 kV dimensions specified and approved by the author. Claude (Anthropic) was used as a coding assistant to write parts of that toolchain and the included C# scripts, under the author's direction and review. No generative AI was used to create meshes, textures, materials, documentation images or marketing images; all textures are procedural and were generated by the author's own code.

No third-party assets, fonts or code are included.